

SCIENCE PRO

CLASS 10 · BIOLOGY · CHAPTER 1

Genetics of Life — PYQs

This is a SAMPLE set of exam-style questions, written to demonstrate the download feature. Replace this file with your actual previous year question paper via VSCode — keep the same filename so the site picks it up automatically.

Section A — One-mark questions

- 1 Define a gene.
- 2 Name the two bases that pair with each other in DNA, besides A and T.
- 3 What is the shape of a DNA molecule called?
- 4 Where in the cell is DNA mainly found?

Section B — Two-mark questions

- 1 Distinguish between a gene and a chromosome.
- 2 What is chromatin, and how is it related to a chromosome?
- 3 State the base-pairing rule that applies to DNA.

Section C — Short answer

- 1 Explain, in order, how a gene, a chromosome, DNA, and a cell's nucleus are related to one another.
- 2 How many chromosomes are present in a normal human cell? How are they arranged?
- 3 Why is the fixed base-pairing rule (A–T, G–C) important for DNA to copy itself accurately?

Section D — Diagram-based

- 1 Draw a simple labelled diagram of the DNA double helix, showing the two strands and at least two base pairs.
- 2 Draw a labelled diagram of a chromosome, marking the centromere.